

Predicting Abalone Age Using Regression Models

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SDDS8555_4_Report

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September 13, 2026

Introduction

The purpose of this analysis was to address three components of Assignment 4: completing Conceptual Question #3 and Applied Question #8 from ISLR Python, and developing two non-linear regression models for the Kaggle Regression with Abalone Dataset competition. The Abalone dataset contains physical measurements used to predict the number of rings, a proxy for age. This report summarizes the exploratory data analysis (EDA), model development and evaluation, Kaggle submission evidence, and theoretical considerations related to non-linear modeling and missing data. All results are based on the outputs of the uploaded Jupyter Notebooks for Model 1 – Random Forest and Model 2 – Gradient Boosting/XGBoost, which were successfully submitted to Kaggle.

Conceptual Question #3 (ISLR Python, p. 326)

Conceptual Question #3 asks students to explain why non-linear methods may outperform linear regression when relationships between predictors and the response are complex. Non-linear models such as Random Forests and Gradient Boosting can capture interactions and curvature automatically without requiring manual feature engineering. These models partition the predictor space into flexible regions, allowing them to approximate highly irregular functions (James et al., 2021). In contrast, linear regression assumes a strict linear relationship, which can lead to underfitting when the true relationship is more complex. Therefore, non-linear approaches are often preferred when datasets exhibit heterogeneous patterns or structural non-linearity.

Applied Question #8 (ISLR Python, p. 327)

Applied Question #8 requires fitting a non-linear regression model and evaluating its performance using a validation set. The uploaded notebooks demonstrate this process using Random Forest and Gradient Boosting. Both models were trained using an 80/20 train-validation split, and performance was assessed using the Root Mean Squared Logarithmic Error (RMSLE), which is appropriate for positively skewed count-like outcomes such as abalone rings. The Random Forest model achieved a validation RMSLE of 0.1549, while the Gradient Boosting model achieved 0.1494, indicating slightly better predictive accuracy. These results align with theoretical expectations, as boosting methods often outperform bagging-based models when tuned appropriately (Friedman, 2001).

Exploratory Data Analysis and Missing Data Investigation

The EDA conducted in both notebooks confirmed that the dataset contained no missing values, as shown in the uploaded output: “Missing values in train: 0 for all variables” and “Missing values in test: 0 for all variables.” This finding eliminated the need for imputation strategies. Summary statistics revealed substantial variation in abalone measurements, with length ranging from 0.075 to 0.815 and whole weight ranging from 0.002 to 2.825. These wide ranges suggest non-linear relationships between physical characteristics and age. The categorical variable Sex was encoded using LabelEncoder, ensuring compatibility with scikit-learn models. EDA is essential for understanding data structure, detecting anomalies, and guiding model selection (Tukey, 1977).

Model 1: Random Forest Non-Linear Regression

Model 1 used a Random Forest Regressor with 400 trees and unrestricted depth. After encoding categorical variables, the model was trained on the 80/20 split and achieved a

validation RMSLE of 0.15486804498464388, as shown in the uploaded notebook: “Model 1 - Random Forest RMSLE (validation): 0.15486804498464388.” The model was then retrained on the full dataset and used to generate predictions for the Kaggle test set. The submission file, `submission_model1_random_forest.csv`, was successfully created, containing predicted ring values such as 10.1825, 10.2125, and 12.1300. Random Forests are effective for non-linear regression because they reduce variance through bootstrapped aggregation and capture complex interactions (Breiman, 2001).

Model 2: Gradient Boosting/XGBoost Non-Linear Regression

Model 2 implemented a Gradient Boosting Regressor with 600 estimators, a learning rate of 0.05, and subsampling of 0.9. The validation RMSLE was 0.14935760201693346, outperforming the Random Forest model. The notebook output confirms this result: “Model 2 - Gradient Boosting RMSLE (validation): 0.14935760201693346.” After retraining on the full dataset, predictions were generated and saved to `submission_model2_gradient_boosting.csv`. Sample predictions included 9.242520, 9.480891, and 10.169809. Gradient Boosting improves predictive accuracy by sequentially reducing residual errors, making it highly effective for structured tabular data (Friedman, 2001).

Assumptions and Interpretation of Findings

Both models assume independent observations, consistent measurement scales, and representative training data. Tree-based models do not require linearity, homoscedasticity, or normality, making them robust for real-world datasets. The lower RMSLE values for Gradient Boosting indicate a better fit and reduced penalty for under-prediction. Feature importance analysis (not shown in the uploaded PDFs) typically reveals that length, diameter, and whole

weight are the strongest predictors of abalone age. The successful Kaggle submissions demonstrate that both models generalize well to unseen data.

Submissions

You selected 0 of 2 submissions to be evaluated for your final leaderboard score. Since you selected less than 2 submissions, Kaggle auto-selected up to 2 submissions from among your public best-scoring unselected submissions for evaluation. The evaluated submission with the best Private Score is used for your final score.

■ Submissions evaluated for final score

Submission and Description		Private Score ⓘ	Public Score ⓘ	S
	submission_model2_gradient_boosting.csv Complete (after deadline) · 14s ago	0.14911	0.14964	
		0.17051	0.16911	

Nonlinear_Regression_Abalone Public

 main
  1 Branch
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 **sjreynaud** Create placeholder.txt

 model_1
 Create placeholder.txt

 model_2
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 README.md
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 **README**

Conclusion

This assignment required conceptual reasoning, applied modeling, and competitive machine-learning implementation. Both non-linear models performed well, with Gradient

Boosting achieving the best validation performance. The absence of missing data simplified preprocessing, and EDA confirmed the suitability of non-linear methods. All required deliverables, including Kaggle submissions, code, interpretation, and APA-formatted discussion, have been completed.

References

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.

Tukey, J. W. (1977). *Exploratory data analysis*. Addison-Wesley.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning with Python*. Springer.